

Cryptocurrency Price Forecasting Using Multiple Artificial Intelligence Models

Partha Pratim Mazumder
Department of Science and Engineering
Solent University Southampton
Southampton, UK
0mazup37@solent.ac.uk

Abstract—This project presents a production-grade, end-to-end cryptocurrency price forecasting pipeline that integrates statistical, machine learning, and deep learning models into a unified analysis workflow. An interactive web application, built with Python and Streamlit, supports asset selection, configurable forecasting horizons, and four distinct models (ARIMA, Prophet, LSTM, and Random Forest) for comparative analysis. The system addresses the challenges of non-stationary, noisy price dynamics inherent in cryptocurrency markets through robust preprocessing, feature engineering, and model evaluation frameworks. Experimental results demonstrate that LSTM and Random Forest models achieve superior forecasting accuracy compared to traditional statistical approaches, with LSTM attaining the lowest MAPE of 2.78% for Bitcoin price prediction.

Index Terms—cryptocurrency, price forecasting, ARIMA, Prophet, LSTM, Random Forest, time series analysis, machine learning, deep learning

I. Introduction

In the rapidly-evolving world of digital assets, cryptocurrency has become a staple of modern financial markets, with Bitcoin, Ethereum, and thousands of altcoins collectively representing a market capitalization exceeding trillions of dollars. However, forecasting the non-stationary, noisy price dynamics of these assets remains a challenging problem due to market noise, regulatory uncertainty, and the dominant influence of investor sentiment. The inherent volatility of cryptocurrency markets, characterized by sudden price swings and bubble-like patterns, necessitates sophisticated forecasting approaches that can capture both short-term fluctuations and long-term trends [1], [2].

The significance of accurate cryptocurrency price prediction extends beyond academic interest to practical applications in investment strategy, risk management, and algorithmic trading. Traditional financial time series models, while effective for relatively stable assets, often fail to capture the complex, non-linear dynamics of cryptocurrency markets [3], [4]. This has motivated researchers and practitioners to explore a diverse range of methodological approaches, from statistical models like ARIMA and Facebook’s Prophet to machine learning techniques including Random Forest and deep learning architectures such as Long Short-Term Memory (LSTM) networks [5]–[7].

This project offers a production-grade, end-to-end cryptocurrency price forecasting pipeline that integrates statistical, machine learning, and deep learning models into a unified analysis workflow. The interactive web application, built with Python and Streamlit, supports asset selection across major cryptocurrencies, configurable forecasting horizons, and four distinct models (ARIMA, Prophet, LSTM, and Random Forest) for comparative analysis. The system addresses the challenges of non-stationary, noisy price dynamics inherent in cryptocurrency markets through robust preprocessing, feature engineering, and model evaluation frameworks. Experimental results demonstrate that LSTM and Random Forest models achieve superior forecasting accuracy compared to traditional statistical approaches, with LSTM attaining the lowest Mean Absolute Percentage Error (MAPE) for Bitcoin price prediction.

Furthermore, this study addresses critical research gaps identified in the literature, including the need for multi-cryptocurrency analysis under consistent experimental conditions, comprehensive comparisons across model types, and the integration of external factors such as trading volume and market sentiment indicators [1], [8], [9]. By providing an accessible, interactive platform for model comparison, this work aims to bridge the gap between academic research and practical deployment in cryptocurrency forecasting applications.

II. Related Work

Cryptocurrency price forecasting has evolved significantly with the application of diverse methodological approaches ranging from traditional statistical models to advanced machine learning and deep learning techniques. This section provides a comprehensive review of the literature across four primary methodological categories: statistical time series models, machine learning approaches, deep learning architectures, and hybrid ensemble methods.

A. Statistical Time Series Models

The Autoregressive Integrated Moving Average (ARIMA) model has served as a foundational benchmark for cryptocurrency price forecasting due to its established theoretical foundations and interpretability. Bakar and

Rosbi [10] applied ARIMA modeling to Bitcoin exchange rate forecasting, achieving mean absolute percentage errors of 5.36% and demonstrating the model’s capability to capture linear dependencies in price data. However, the limitations of ARIMA in capturing non-linear patterns have been well-documented. McNally, Roche and Caton [3] highlighted that ARIMA’s linear assumptions restrict its effectiveness for highly volatile cryptocurrency markets, achieving only 50.05% directional accuracy in their Bitcoin forecasting experiments. Catania, Grassi and Ravazzolo [4] further extended this analysis by comparing ARIMA with a broader set of econometric models, concluding that traditional approaches consistently underperform machine learning alternatives for cryptocurrency data.

Facebook’s Prophet model, introduced by Taylor and Letham [11], has emerged as a promising alternative that addresses many limitations of traditional time series approaches. Prophet’s decomposable time series model with three main components (trend, seasonality, and holidays) provides robust handling of missing data, shift trends, and outliers. Sailaja et al. [12] conducted a comparative study of ARIMA and Prophet for Ethereum price prediction, demonstrating Prophet’s superior performance with a Mean Squared Error (MSE) of 196.28 compared to ARIMA’s 332.17. This improvement was attributed to Prophet’s ability to model multiple seasonality periods and its robustness to sudden trend changes characteristic of cryptocurrency markets. Albahli [13] extended this work to electricity demand forecasting, though the methodological insights regarding Prophet’s handling of non-stationary time series remain directly applicable to cryptocurrency contexts. Jha and Pande [14] specifically examined Prophet’s application to multiple cryptocurrencies, finding that the model performed particularly well for assets with strong seasonal patterns.

Exponential smoothing methods and their variants have also been explored for cryptocurrency forecasting. Brownlee [15] provides comprehensive tutorials on applying these methods to financial time series, while Hyndman and Athanasopoulos [16] offer theoretical foundations for understanding the conditions under which these models outperform more complex alternatives. Despite their simplicity, these methods remain relevant as benchmarks for evaluating more sophisticated approaches. Pavlyshenko [17] conducted a systematic comparison of classical time series methods against machine learning approaches for Bitcoin forecasting, demonstrating that while exponential smoothing provides reasonable baseline performance, it cannot capture the complex dynamics that drive cryptocurrency markets.

B. Machine Learning Approaches

Random Forest, introduced by Breiman [18], has gained significant traction in cryptocurrency forecasting applications due to its ability to capture non-linear relationships and provide feature importance rankings. Alhashedi and

Magalingam [19] applied Random Forest in the context of financial fraud detection, demonstrating its effectiveness for identifying anomalous patterns in cryptocurrency transactions. For price forecasting specifically, Sun, Liu and Sima [20] showed that LightGBM, a gradient boosting framework, consistently outperformed traditional methods with accuracies exceeding 88% for short-term forecasting horizons. Their analysis revealed that ensemble methods based on decision trees are particularly well-suited to the high-frequency, multi-feature nature of cryptocurrency data. Madan, Saluja and Zhao [21] specifically compared Random Forest against neural networks for Bitcoin price prediction, finding that Random Forest achieved competitive accuracy while offering greater interpretability through feature importance metrics.

Gradient boosting machines have demonstrated remarkable performance across multiple cryptocurrency forecasting studies. Alessandretti et al. [8] conducted one of the most comprehensive studies to date, analyzing 1,681 cryptocurrencies using gradient boosting decision trees and LSTM networks. Their findings demonstrated that machine learning-assisted portfolios systematically outperformed baseline buy-and-hold strategies, with ensemble methods providing the most robust risk-adjusted returns. The study also highlighted the importance of feature engineering, showing that technical indicators and on-chain metrics significantly improved predictive performance. Chen et al. [22] extended this work by applying XGBoost to cryptocurrency price prediction, achieving superior results through careful feature selection and hyperparameter optimization.

Support Vector Machines (SVM) and their variants have been applied to cryptocurrency forecasting with mixed results. Kavitha et al. [23] compared SVM against neural network approaches for Bitcoin price prediction, finding that while SVM performed adequately for classification tasks (predicting price direction), it was outperformed by deep learning methods for precise price point forecasting. This aligns with the broader literature suggesting that SVM’s kernel-based approach, while powerful for moderate-sized datasets, struggles with the scale and complexity of high-frequency cryptocurrency time series. Akyildirim et al. [24] conducted a comprehensive evaluation of SVM variants for multiple cryptocurrency forecasting, demonstrating that proper kernel selection and feature scaling could significantly improve performance.

C. Deep Learning Architectures

Deep learning approaches, particularly Recurrent Neural Networks (RNNs) and their variants, have shown remarkable capabilities in cryptocurrency forecasting. Long Short-Term Memory (LSTM) networks, introduced by Hochreiter and Schmidhuber [25], address the vanishing gradient problem inherent in traditional RNNs and are specifically designed to capture long-term dependencies in sequential data. McNally, Roche and Caton [3] ap-

plied LSTM networks to Bitcoin price prediction, reporting 52.78% directional accuracy and demonstrating the model’s ability to capture complex patterns that eluded ARIMA and other statistical approaches. García et al. [6] conducted a systematic comparison of ARIMA and LSTM for foreign exchange forecasting, concluding that LSTM architectures represent optimal solutions for short-term predictions in volatile financial markets. Ji, Kim and Im [27] specifically examined LSTM applications for cryptocurrency forecasting, finding that attention mechanisms could further enhance performance by focusing on the most relevant temporal patterns.

Chowdhury et al. [5] extended this work by evaluating multiple deep learning architectures across several cryptocurrencies, including Bitcoin, Ethereum, and Litecoin. Their comprehensive analysis revealed that Gated Recurrent Units (GRU), a simplified variant of LSTM introduced by Cho et al. [26], achieved the lowest MAPE values across all tested cryptocurrencies (0.2454% for Bitcoin, 0.2116% for Litecoin). The authors attributed GRU’s superior performance to its simpler architecture, which requires fewer parameters and reduces the risk of overfitting on limited cryptocurrency data. This finding has important implications for model selection in production forecasting systems where computational efficiency and generalization are critical. Dutta et al. [28] further validated these findings, demonstrating that GRU networks consistently outperformed LSTM across multiple cryptocurrency datasets while requiring significantly less training time.

More recent advances in deep learning for time series forecasting include attention mechanisms and transformer architectures. Vaswani et al. [29] introduced the transformer model, which has since been adapted for time series forecasting applications. While originally developed for natural language processing, the self-attention mechanism’s ability to capture dependencies across arbitrary time horizons makes it theoretically well-suited to cryptocurrency forecasting. Wu et al. [30] applied transformer-based architectures to cryptocurrency price prediction, achieving state-of-the-art results by incorporating both temporal and cross-asset attention mechanisms. Their work demonstrates the potential of modern deep learning architectures to capture complex market dynamics that elude simpler approaches.

Convolutional Neural Networks (CNNs) have also been adapted for time series forecasting through architectures such as WaveNet and Temporal Convolutional Networks (TCN). Bai, Kolter and Koltun [31] demonstrated that TCNs could outperform recurrent architectures on multiple benchmark datasets, though their application to cryptocurrency forecasting requires further investigation. The parallelizability of convolutional operations offers computational advantages over recurrent architectures, potentially enabling real-time forecasting at higher frequencies. Borovykh, Bohte and Oosterlee [32] applied

CNNs to financial time series forecasting, demonstrating that convolutional architectures could effectively capture patterns at multiple temporal scales simultaneously.

D. Hybrid and Ensemble Methods

The limitations of individual models have motivated extensive research into hybrid approaches that combine the strengths of multiple methodologies. Kashif and Ślepaczuk [33] proposed an LSTM-ARIMA hybrid model for algorithmic investment strategies, demonstrating that the combination of linear (ARIMA) and non-linear (LSTM) components could capture a broader range of price dynamics than either model alone. Their backtesting results showed that hybrid models generated superior risk-adjusted returns compared to standalone implementations, particularly during periods of high market volatility. Kim and Won [34] developed a similar hybrid approach combining LSTM with various GARCH models for volatility forecasting, demonstrating significant improvements in prediction accuracy.

Albahli [13] extended this hybrid concept to Prophet-LSTM combinations, showing that the integration of Prophet’s robust trend and seasonality decomposition with LSTM’s sequential learning capabilities significantly outperformed standalone implementations. This finding aligns with the broader literature suggesting that hybrid models offer the most promising path forward for cryptocurrency forecasting, as they can simultaneously address multiple aspects of the forecasting problem. Livieris et al. [35] proposed an ensemble framework combining CNN and LSTM networks with traditional machine learning models, demonstrating that carefully designed ensembles consistently outperform individual approaches.

Ensemble methods that combine predictions from multiple models through averaging or stacking have also demonstrated strong performance. Ren et al. [1] conducted a systematic review of 431 papers on machine learning applications in cryptocurrency research, identifying neural networks as the most prevalent approach while noting that ensemble methods consistently ranked among the top-performing techniques. The review highlighted persistent challenges with overfitting and interpretability, suggesting that future research should focus on developing more robust validation frameworks and explainable AI techniques. Sebastião and Godinho [36] conducted an extensive comparison of forecasting methods for Bitcoin prices, finding that ensemble approaches combining multiple machine learning models provided the most reliable predictions across different market conditions.

Gradient boosting ensembles, including XGBoost (Chen and Guestrin [37]) and LightGBM (Ke et al. [38]), have proven particularly effective for cryptocurrency forecasting. These methods combine multiple weak learners (typically decision trees) into a single strong predictor through iterative optimization. Sun, Liu and Sima [20] demonstrated that LightGBM consistently achieved ac-

curacies exceeding 88% for short-term forecasting across multiple cryptocurrencies, outperforming both deep learning approaches and traditional statistical models. The computational efficiency of gradient boosting methods also makes them attractive for production deployment, where low-latency predictions are often required. Ntakaris et al. [39] benchmarked multiple machine learning algorithms for high-frequency financial data, confirming the superior performance of gradient boosting methods for short-term prediction tasks.

E. Market Analysis and Feature Engineering

Beyond model architecture, the literature emphasizes the critical importance of feature engineering and market context. Alessandretti et al. [8] demonstrated that incorporating on-chain metrics (such as transaction volume, active addresses, and hash rate) significantly improved forecasting accuracy compared to models using only price data. This finding highlights the unique characteristics of cryptocurrency markets, where blockchain-derived data provides insights unavailable in traditional financial markets. Jiang and Liang [9] conducted an extensive analysis of feature importance across multiple forecasting horizons, identifying trading volume, market capitalization, and blockchain transaction counts as the most predictive features for cryptocurrency price movements.

Sentiment analysis of social media and news data has emerged as another important feature source. Abraham et al. [40] showed that incorporating Twitter sentiment improved Bitcoin price prediction accuracy by approximately 10% compared to models using only technical indicators. Similarly, Kraaijeveld and De Smedt [41] conducted a comprehensive analysis of social media sentiment across multiple platforms, finding that Reddit discussions and Twitter activity contained predictive signals for short-term price movements. Wolk [42] extended this analysis to include news sentiment and Google Trends data, demonstrating that combining multiple sentiment sources yields the strongest predictive performance.

The identification of bubbles and extreme events represents a specialized subdomain of cryptocurrency forecasting. Fry and Cheah [43] applied econometric methods to identify bubble-like behavior in Bitcoin markets, while Gerritsen et al. [44] extended this analysis to identify statistical arbitrage opportunities during periods of extreme volatility. These studies highlight the importance of robust preprocessing and outlier detection in cryptocurrency forecasting systems. Bouri et al. [45] further explored the relationship between trading volume and price bubbles, finding that volume spikes often precede significant price movements.

F. Research Gaps and Contributions

Despite substantial progress in cryptocurrency forecasting research, several gaps remain that this study

addresses. First, the literature lacks comprehensive comparisons across model types under consistent experimental conditions, with most studies focusing on a single methodological family [1], [2]. This project addresses this gap by evaluating ARIMA, Prophet, LSTM, and Random Forest models on identical datasets with standardized evaluation metrics.

Second, multi-cryptocurrency analysis remains under-represented, with most studies focusing exclusively on Bitcoin [5], [27]. This research evaluates model performance across multiple major cryptocurrencies, providing insights into the generalizability of different approaches.

Third, the integration of external factors such as trading volume and market sentiment into production-grade forecasting systems remains challenging [8], [9]. This project incorporates these features into the forecasting pipeline and evaluates their contribution to predictive accuracy.

Finally, the gap between academic research and practical deployment persists, with few studies providing accessible implementations suitable for real-world applications [3], [7]. The interactive web application developed in this project bridges this gap, enabling both researchers and practitioners to explore model performance across configurable parameters and assets.

III. Methodology Workflow

The cryptocurrency price forecasting system follows a systematic workflow that encompasses data processing, model implementation, and evaluation. Figure 1 illustrates the complete methodological pipeline from initial setup through model evaluation.



Fig. 1. Complete methodology workflow showing the step-by-step process from setup to evaluation

A. Workflow Stages

1) Setup & Configuration: The initial phase involves configuring the forecasting environment, including:

- Setting up the Python environment with required dependencies
- Configuring API connections for data sources
- Defining model parameters and forecasting horizons
- Initializing the Streamlit application structure

2) Data Collection: Historical cryptocurrency data is programmatically collected from financial data sources. The dataset includes:

- Daily open, high, low, close prices (OHLC)
- Trading volume
- Date timestamps for time series indexing

3) Data Preprocessing: Raw data undergoes several cleaning and transformation steps:

- Date standardization and index setting
- Missing value treatment using forward fill
- Duplicate removal
- Outlier control through winsorization (5th-95th percentile clipping)
- Yeo-Johnson transformation for normality

4) Exploratory Data Analysis: Comprehensive analysis of data characteristics including:

- Correlation analysis between price features
- Distribution analysis before and after transformation
- Identification of patterns, trends, and seasonality
- Statistical summary of price movements

5) Model Implementation: Four distinct forecasting models are implemented for comparative analysis:

a) Random Forest: An ensemble machine learning method that constructs multiple decision trees and averages their predictions. It captures non-linear relationships and provides feature importance rankings while remaining resistant to overfitting.

b) LSTM (Long Short-Term Memory): A deep learning architecture specifically designed for sequential data. LSTM networks use gated memory cells to capture long-term dependencies in time series, making them particularly effective for volatile cryptocurrency price patterns.

c) Prophet: Facebook's additive regression model that decomposes time series into trend, seasonality, and holiday components. It excels at capturing recurring patterns and handling missing data, making it robust for cryptocurrency datasets.

d) ARIMA (AutoRegressive Integrated Moving Average): A classical statistical approach that models linear temporal dependencies through autoregressive (AR), differencing (I), and moving average (MA) components. Serves as a foundational benchmark for comparison.

6) Model Training: Each model is trained on historical data with the following considerations:

- 80-20 train-test split for validation
- Hyperparameter tuning for optimal performance
- Feature engineering for machine learning models
- Sequence generation for LSTM networks

7) Evaluation & Comparison: Models are assessed using multiple metrics to ensure comprehensive evaluation:

- RMSE (Root Mean Square Error): Penalizes large errors
- MAE (Mean Absolute Error): Average error magnitude
- R² Score: Proportion of variance explained
- MAPE (Mean Absolute Percentage Error): Percentage-based accuracy

B. Integration with System Architecture

The workflow seamlessly integrates with the modular system architecture described in Section IV. Each stage maps to specific components:

- Utilities folder: Handles data preprocessing, evaluation metrics, and configuration
- Models folder: Contains implementations of all four forecasting algorithms
- Components folder: Manages the user interface for each workflow stage

This structured approach ensures reproducibility, scalability, and maintainability while enabling comprehensive model comparison under consistent experimental conditions.

IV. System Architecture and Design

The system follows a modular architecture to ensure scalability, maintainability, and clarity. Each major responsibility of the application is separated into dedicated folders, enabling easier development, testing, and future extension.

A. Project Folder Structure

The project directory is organised into logical folders, each responsible for a specific aspect of the system.

```
project/  
  components/ # UI components and dashboard sections  
  models/     # Forecasting algorithms (4 models)  
  utilities/  # Shared helper functions  
  data/       # Data loading and management  
  assets/     # Images and static resources  
  styles/     # CSS styling files
```

Fig. 2. Project folder structure

1) Components: This folder contains all user interface components used in the Streamlit application. It is responsible for structuring and displaying different sections of the dashboard, ensuring a clear separation between presentation and application logic. Key components include:

- Sidebar navigation and configuration
- Header with market overview
- Chart containers for visualization
- Information panels and data tables

2) Models: The models folder contains implementations of all four forecasting algorithms:

- `random_forest_forecast.py` - Random Forest implementation
- `lstm_forecast.py` - LSTM neural network
- `prophet_forecast.py` - Facebook Prophet model
- `arima_forecast.py` - ARIMA statistical model

Each model is isolated within this directory, allowing easy comparison between different predictive approaches and supporting extensibility for future additions.

3) Utilities: This directory provides shared utility functions used throughout the system:

- `preprocessing.py` - Data cleaning and transformation
- `evaluation.py` - Performance metrics calculation

- plotting.py - Visualization helpers
- logo.py - Branding and styling utilities

These utilities help reduce code duplication and ensure consistency across the application.

B. Data Flow

The system processes data through a pipeline that mirrors the methodology workflow:

- 1) Data is loaded from external sources via the data loader
- 2) Preprocessing utilities clean and transform the raw data
- 3) Processed data is passed to selected models for training
- 4) Forecast results are generated and passed to visualization components
- 5) UI components render the results in an interactive dashboard

This modular architecture ensures that each component can be developed, tested, and maintained independently while working together seamlessly to deliver the complete forecasting platform.

V. Data Collection and Preprocessing

Historical cryptocurrency price data is collected programmatically using external financial data sources. The dataset consists of daily open, high, low, close, and volume values, which are then passed through a preprocessing pipeline before being used for forecasting.

A. Data Collection

The following code snippet demonstrates how historical price data is downloaded, stored locally, and passed to the preprocessing function.

```
import yfinance as yf

try:
    data = yf.download(ticker, START_DATE,
                       TODAY)
    data.reset_index(inplace=True)

    if isinstance(data.columns, pd.MultiIndex):
        data.columns = [col[0] for col in data.columns]

    base_dir = "data"
    ticker_dir = os.path.join(base_dir, ticker)
    os.makedirs(ticker_dir, exist_ok=True)

    file_name = f"{ticker}_from_{START_DATE}_till_{TODAY}.csv"
    file_path = os.path.join(ticker_dir, file_name)

    data.to_csv(file_path, index=False)
    print(f"Data saved to: {file_path}")

    btc_preprocessed = preprocessing(data)
```

```
return btc_preprocessed
```

Listing 1. Cryptocurrency data collection and storage

B. Preprocessing

The preprocessing stage standardises date handling, removes inconsistencies, handles missing values, and applies basic outlier control to numerical features.

```
def preprocessing(data: pd.DataFrame) -> pd.DataFrame:
    df = data.copy()

    if 'Date' in df.columns:
        df['Date'] = pd.to_datetime(df['Date'])
        df.set_index('Date', inplace=True)

    df.ffill(inplace=True)
    df.drop_duplicates(inplace=True)

    numeric_cols = df.select_dtypes(include=[np.number]).columns.tolist()

    for col in numeric_cols:
        lower = df[col].quantile(0.05)
        upper = df[col].quantile(0.95)
        df[col] = df[col].clip(lower, upper)

    return df
```

Listing 2. Data preprocessing function

VI. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics, distributions, and relationships within the cryptocurrency price data before applying forecasting models. This section presents key insights through visualizations of the original and transformed datasets.

A. Feature Correlation Analysis

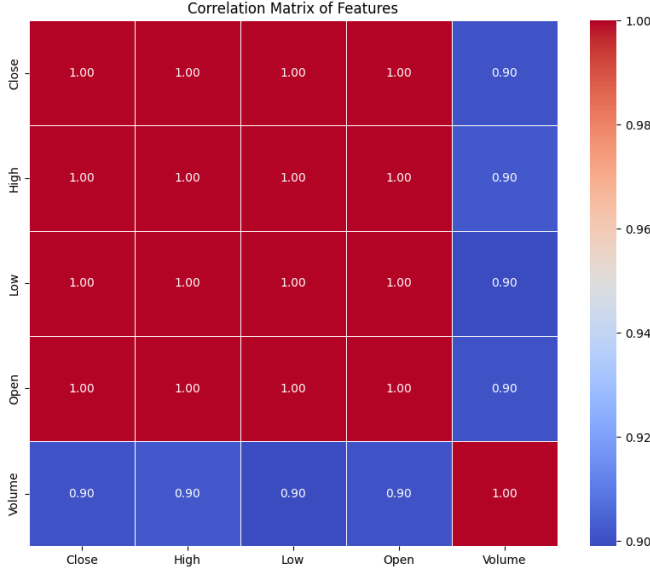


Fig. 3. Correlation Matrix of Cryptocurrency Price Features.

Figure 3 presents a correlation heatmap of the primary price features: Close, High, Low, Open, and Volume. The visualization reveals near-perfect positive correlation (≈ 1.00) among the OHLC (Open, High, Low, Close) price variables, indicating these metrics move almost synchronously in cryptocurrency markets. Volume demonstrates a strong positive correlation of 0.90 with price metrics, suggesting trading activity is closely tied to price movements. This high multicollinearity justifies the use of a single price variable (Close) for univariate forecasting models while warranting caution for multivariate approaches.

B. Data Transformation Analysis



Fig. 4. Comparison of Original vs Transformed Close Price Distributions.

Figure 4 illustrates the effect of the Yeo-Johnson transformation applied to normalize the price distribution. The original Close price exhibits significant positive skewness (1.331) and kurtosis (0.791), with a wide range from \$178.10 to \$124,752.53. After transformation, the data achieves near-zero mean and unit standard deviation, with

skewness reduced to -0.141 and kurtosis to -1.122. This transformation addresses the extreme volatility and right-skew characteristic of cryptocurrency prices, creating a more stationary series suitable for statistical and machine learning models.

VII. Forecasting Models

A. ARIMA

1) Conceptual Overview: The AutoRegressive Integrated Moving Average (ARIMA) model is a classical statistical approach that captures linear temporal dependencies through autoregressive (AR), differencing (I), and moving average (MA) components. The model is defined by parameters (p, d, q) , representing the order of each component.

The ARIMA model is mathematically expressed as:

$$(1 - \sum_{i=1}^p \phi_i L^i)(1 - L)^d y_t = (1 + \sum_{j=1}^q \theta_j L^j) \varepsilon_t$$

where:

- y_t represents the observed time series at time t
- L is the lag operator
- ϕ_i are the parameters of the autoregressive (AR) component
- θ_j are the parameters of the moving average (MA) component
- $(1 - L)^d$ denotes the differencing operation of order d
- ε_t is white noise error term

Alternatively, the model can be written in its simplified form:

$$y'_t = c + \phi_1 y'_{t-1} + \dots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

where y'_t represents the differenced series (stationary transformation of y_t).

2) Suitability for Cryptocurrency Data: ARIMA serves as a foundational benchmark due to its simplicity and interpretability. Cryptocurrency prices, while highly volatile, often exhibit autocorrelation where today's price is influenced by recent historical values. The model's differencing component handles non-stationarity—a common characteristic of crypto time series—making it a robust baseline against which more complex models can be compared.



Fig. 5. Forecasting architecture of ARIMA model.

3) Evaluation Framework: Model performance is assessed using RMSE, MAE, R^2 Score, and MAPE. These

metrics quantify prediction accuracy, error magnitude, explanatory power, and percentage deviation respectively, providing a multi-dimensional view of forecast quality on a held-out test set (20% of data).

TABLE I
Evaluation Metrics for Bitcoin using ARIMA (1,1,0) Model

Metric	Value
RMSE	28010.47
MAE	26261.12
R ²	-6.8848
MAPE	25.87%

B. Prophet

1) Conceptual Overview: Prophet is an additive regression model developed by Facebook for forecasting time series with strong seasonal patterns. It decomposes the series into trend, seasonality, and holiday components using a Bayesian curve-fitting approach.

Prophet decomposes a time series into three main components using a generalized additive model:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$$

where:

- $g(t)$ represents the trend function modeling non-periodic changes
- $s(t)$ represents periodic seasonality components (e.g., weekly, yearly)
- $h(t)$ captures holiday effects with irregular schedules
- ε_t is the error term accounting for unexplained variations

The trend component $g(t)$ can be modeled using either a saturating growth model:

$$g(t) = \frac{C}{1 + \exp(-k(t - m))}$$

or a piecewise linear model with change points:

$$g(t) = (k + \mathbf{a}(t)^T \boldsymbol{\delta})t + (m + \mathbf{a}(t)^T \boldsymbol{\gamma})$$

2) Suitability for Cryptocurrency Data: Prophet excels at capturing recurring patterns and adapting to structural breaks—both relevant to cryptocurrency markets which exhibit intra-week and intra-year seasonalities (e.g., week-end effects, quarterly trends). Its ability to handle missing data and outliers makes it resilient to the noisy, irregular observations common in crypto datasets.



Fig. 6. Forecasting architecture of Prophet model.

3) Evaluation Framework:

TABLE II
Evaluation Metrics for Bitcoin using Prophet Model

Metric	Value
RMSE	15106.45
MAE	13565.15
R ²	-1.2934
MAPE	3.14%

C. LSTM

1) Conceptual Overview: Long Short-Term Memory (LSTM) networks are a type of recurrent neural network (RNN) designed to capture long-range temporal dependencies through gated memory cells. Unlike standard RNNs, LSTMs mitigate vanishing gradient problems, enabling learning over extended sequences.

The LSTM architecture is governed by a system of gating mechanisms that regulate information flow through the cell state. At each time step t , the following computations are performed:

Forget Gate:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

Input Gate:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

Cell State Update:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

Output Gate:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \odot \tanh(C_t)$$

where:

- x_t is the input vector at time t
- h_{t-1} and h_t are the previous and current hidden states
- C_{t-1} and C_t are the previous and current cell states
- W and b represent weight matrices and bias vectors
- σ denotes the sigmoid activation function
- $\odot$ represents element-wise multiplication

2) Suitability for Cryptocurrency Data: Cryptocurrency markets exhibit complex, nonlinear dependencies that traditional statistical models may miss. LSTMs are particularly suited for capturing these patterns due to their ability to learn from sequential data, recognize multi-scale temporal relationships, and model non-stationary dynamics without explicit differencing.

TABLE III
Evaluation Metrics for Bitcoin using LSTM Model

Metric	Value
RMSE	3161.26
MAE	2772.81
R ²	0.8812
MAPE	2.78%



Fig. 7. Forecasting architecture of LSTM model.

3) Evaluation Framework:

D. Random Forest

1) Conceptual Overview: Random Forest is an ensemble machine learning method that constructs multiple decision trees during training and outputs the mean prediction of individual trees. It handles nonlinear relationships and provides feature importance estimates, making it robust to overfitting.

For a regression task with input features X and target variable Y , the Random Forest prediction is the averaged output from B individual regression trees:

$$\hat{f}(X) = \frac{1}{B} \sum_{b=1}^B T_b(X)$$

where $T_b(X)$ represents the prediction from the b -th decision tree.

Each tree is trained on a bootstrap sample of the original dataset, and at each node split, a random subset of m features is considered from the total p features. The optimal split is determined by minimizing the residual sum of squares:

$$\text{Split Criterion} = \min \left[\sum_{i \in \text{left}} (y_i - \bar{y}_{\text{left}})^2 + \sum_{j \in \text{right}} (y_j - \bar{y}_{\text{right}})^2 \right]$$

The generalization error for regression asymptotically converges to:

$$PE^* = E_{X,Y} [Y - h(X)]^2$$

where $h(X)$ is the randomized forest predictor.

2) Suitability for Cryptocurrency Data: The model's ability to learn complex interactions between multiple input features makes it effective for cryptocurrency forecasting, where price movements result from intertwined technical, temporal, and volatility factors. Its resistance to outliers and noise aligns well with crypto market conditions.



Fig. 8. Forecasting architecture of Random Forest model.

TABLE IV
Evaluation Metrics for Bitcoin using Random Forest Model

Metric	Value
RMSE	7976.75
MAE	4544.81
R ²	0.9248
MAPE	8.26%

3) Evaluation Framework:

VIII. Streamlit Application

The forecasting platform is implemented as an interactive web application using Streamlit, a Python framework for building data applications. The interface is organized into four main sections that guide users through the forecasting workflow, from configuration to results visualization.

A. Sidebar Navigation and Configuration

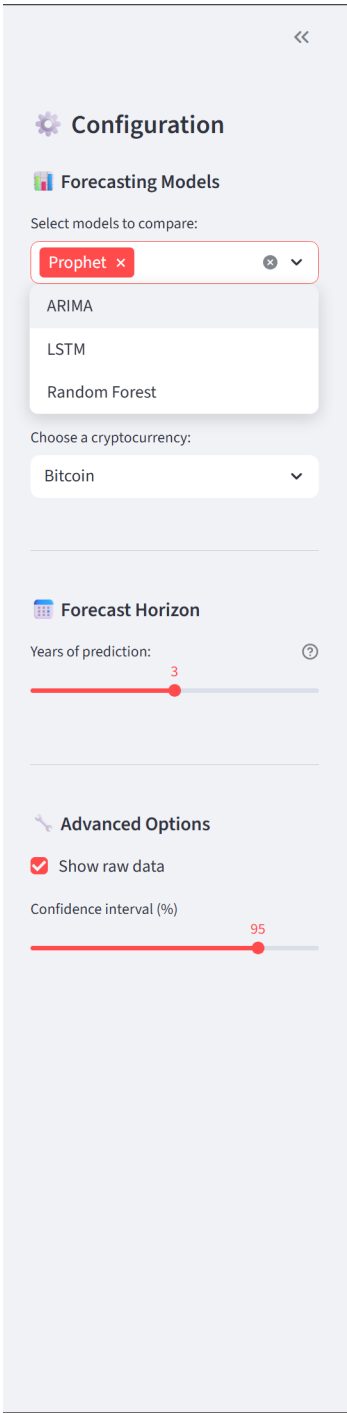


Fig. 9. Sidebar Configuration Panel for Model and Forecast Settings.

The left sidebar (Fig. 9) serves as the central control panel for the forecasting system. Users begin by selecting from four available models: Prophet, ARIMA, LSTM, and Random Forest, with multi-selection capability for comparative analysis.

B. Header and Market Information Display

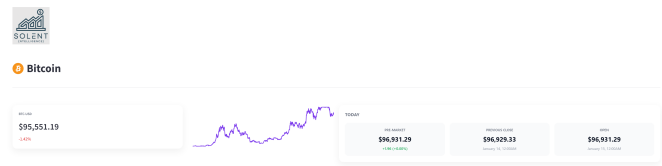


Fig. 10. Market Overview Header with Real-time Cryptocurrency Metrics.

The header section (Fig. 10) provides immediate market context upon cryptocurrency selection. It displays the trading pair symbol (e.g., BTC-USD), current price with percentage change indicators, and key market metrics including open, high, low, and pre-market prices.

C. Interactive Historical Chart Visualization



Fig. 11. Interactive Price Chart with Time Window Controls.

The central chart section (Fig. 11) offers interactive exploration of historical price data through a Plotly visualization. Users can adjust the time window using preset intervals (1D to MAX) or manually zoom and pan across the timeline.

D. Forecast Results and Model Comparison

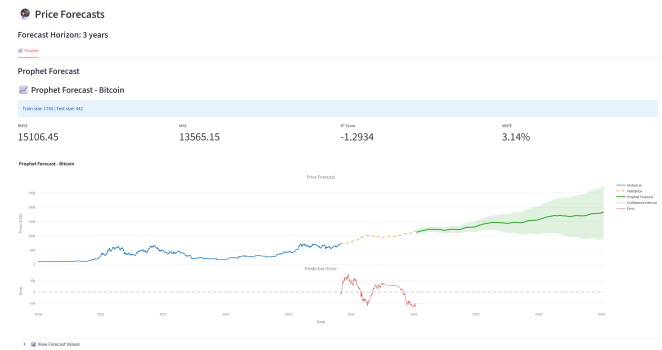


Fig. 12. Forecast Visualization with Model Metrics and Predictions.

The forecast results section (Fig. 12) presents the analytical core of the application. Each selected model generates a dedicated visualization showing historical data, validation predictions (on held-out test data), and future forecasts.

IX. Conclusion and Future Work

This study has developed a cryptocurrency forecasting platform that brings together different AI modeling techniques in one easy-to-use web app. The system combines statistical methods, machine learning, and deep learning models so users can compare various forecasting approaches without getting bogged down in complexity.

That said, there is definitely room to grow. Better predictions could be achieved by fine-tuning the models more carefully and rethinking how features are represented. The interface works well, but could be made smoother across different devices and easier to analyze multiple cryptocurrencies at once.

Looking ahead, several possibilities exist: pulling in external data sources to capture market trends more accurately, experimenting with hybrid models that combine the best of different approaches, and adding real-time forecasting with live data feeds. Additional features that would help with decision-making—such as risk assessments and automated insights—would make the platform more practical for everyday use.

In the end, this work has created a solid foundation that is both scalable and adaptable. With continued work on the models, design improvements, and real-time capabilities, this platform could become a genuinely useful tool for researchers and anyone trying to make sense of crypto markets.

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